



continuity tester. Once the wire having a break is detected, signals from the collectors of transistors T2 and T3 (points A and B) have to be applied to the two ends of that wire and a tube has to be run along the wire for detecting the break point. This tube actually forms a capacitor with the test wire. When the tube crosses the break point in the wire, there is a flip in the LED glow indicating wire break.

The circuit is built around a 555 timer IC (IC1), CMOS NAND gate CD4011 (IC2), CMOS NOR gate CD4001 (IC3) and a few discrete components. IC 555 is wired as an astable multivibrator. Its output is fed to the detector circuit built around transistors T2 through T5. Further, IC2 and IC3 are wired as a flip-flop.

The tube detector (2-5cm long) can be easily built using commonly available thin tin foil with some insulation tape between the outer (10mm dia.) and the inner tubes (6mm dia.).

The test wire influences the capacitance of this detector, as the two ends of the test wire (A and B) are connected to the signals from collectors of T2 and T3 (points A and B). Points A and B are actually out of phase with each other and therefore have opposite effects on the capacitance of the tube. If one increases the effective capacitance, the other decreases it due to the potential difference. This tiny change in capacitance eventually affects the duration of the first trigger pulse produced by the combination of resistor R11 and variable capacitor VC1.

Locating the wire break is simple. First of all, tune the circuit by adjusting trimming capacitor VC1 such that both the green and red LEDs (LED1 and LED2, respectively) glow when there is no wire inside the tube detector capacitor. Then insert the wire into the detector capacitor and connect its A and B ends to the respective points

from collectors of transistor T2 and T3. At this time, only the red LED glows.

Now run the tube detector along the wire to detect the break. When the tube detector crosses the wire break, the green LED too glows. Mark this point of the wire. Note that a bare wire inside the tube increases the capacitance of the detector.

The detector capacitor should be used as close as possible to the circuit and also to the test wire for maximum capacitive effects. So avoid using a long wire to connect the tube detector capacitor to the circuit. The circuit works off around 5V, while the test wire is driven with 12V supply.

Assemble the circuit on a small general-purpose PCB. For convenient handling, fit the entire PCB assembly inside a glue stick tube with the tube detector capacitor protruding outside like a 'T'. Else, mount the entire PCB including the detector tube on a bench. ●

CALL BELL WITH WELCOME INDICATION

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Here is a simple call bell circuit that displays a welcome message when somebody presses the call bell switch momentarily. The

alphanumeric display can be fitted near the call bell switch.

The circuit is built around two 555 ICs (IC1 and IC2), seven KLA511 common-anode alphanumeric displays (DIS1 through DIS7) and a few discrete

components. For easy understanding, the entire circuit can be divided into two sections: controller and display. The controller section is built around IC1 and IC2, while the display section is built around alphanumeric displays

